PSPP Studio for Android — Complete User Guide

GNU PSPP 2.1.2 Native Statistical Engine (org.pspp.mobile) • Version 1.0.0

1. Overview & Architecture

PSPP Studio is a native Android statistical analysis environment that embeds the real GNU PSPP 2.1.2 engine (libpspp, libpspp-core, GSL 2.8, libxml2, gettext/libintl) via JNI (libpspp_jni.so). All syntax execution, statistical procedures, and system file (.sav) processing run locally on your device with full offline support.

Key Capabilities

- Syntax Editor: Highlighted GNU PSPP (.sps) editor with Run All, Run Selection, inline line-number error markers, and bundled clinical trial sample scripts.
- Data & Variable View: Dual-grid spreadsheet editor for case data and variable dictionary attributes (Name, Type, Width, Decimals, Label, Values, Missing, Measure).
- Analyze Menu (9 Procedures): Point-and-click dialogs that generate and execute editable PSPP syntax for Descriptives, Frequencies, Crosstabs, T-Tests, ANOVA, Correlation, Linear Regression, Chi-Square, and Scale Reliability (Cronbach Alpha).
- Output Viewer: Renders native PSPP HTML pivot tables in WebView with one-tap export to HTML, PDF (via Android Print), Plain Text (.txt), CSV, and Share Sheet.
- Storage Access Framework (SAF): Open and save .sav, .zsav, .por, .csv, .txt, and .sps files from device storage, with interactive CSV delimiter & column-type preview.
- Multilingual Support: Full English, Arabic (RTL), and French UI and compiled GNU gettext (.mo) engine message catalogs.

2. Getting Started & Navigation

The bottom navigation bar (or side Navigation Rail on tablets/foldables) provides six primary workspaces:

- 1. Syntax — Write, load, validate, and execute PSPP (.sps) syntax scripts.
- 2. Data — Inspect and edit cases in Data View or metadata in Variable View.
- 3. Analyze — Configure any of the 9 statistical procedures with live syntax preview.
- 4. Output — View formatted statistical tables, diagnostics, and export/print.
- 5. Files — Open/save datasets (.sav/.csv) and scripts (.sps), or load samples.
- 6. About — Inspect JNI engine status, GPLv3 license, patches, and documentation.

3. Using the Syntax Editor (.sps)

The Syntax screen provides a dedicated code editor tailored to GNU PSPP commands.

Running Scripts

- Run All: Executes the entire syntax script in an isolated background working directory and automatically opens the Output screen when finished.
- Run Selection (Lstart-Lend): Adjust the line selection range using the Selection controls and tap Run Selection to execute only the highlighted command block.
- Cancel Execution: While a long procedure is running, tap Cancel in the top progress banner to send SIGTERM / atomic cancel to the active native engine worker.

Bundled Sample Scripts

Tap the 'Samples' button in the Syntax toolbar to load ready-to-run scripts:

- 1. Descriptives & Frequencies (descriptives.sps) — Summary statistics and frequency distributions on the 20-patient clinical_trial.csv dataset.
- 2. Crosstabs & Independent T-Test (crosstabs.sps) — Contingency tables with Pearson Chi-Square, Independent-Samples T-Test, and One-Way ANOVA.
- 3. Correlation, Regression & Reliability (regression.sps) — Pearson r matrix, Ordinary Least Squares (OLS) multiple linear regression, and Cronbach's Alpha.

Syntax Rules & Error Diagnostics

- Every PSPP command must end with a period (.) terminator.
- Comments begin with an asterisk (*) at the start of a line and end with (.).
- If the editor or native engine detects a syntax error, a red badge appears on the offending line number and the diagnostic message is displayed in the error banner.

Example PSPP Syntax Snippet

```
GET FILE='clinical_trial.sav'.
DESCRIPTIVES VARIABLES=age baseline_bp followup_bp outcome_score
  /STATISTICS=MEAN STDDEV MIN MAX VARIANCE SKEWNESS KURTOSIS SEMEAN.
FREQUENCIES VARIABLES=group gender
  /FORMAT=AVALUE TABLE
  /STATISTICS=MEAN STDDEV MEDIAN MODE MIN MAX.
EXECUTE.
```

4. Data Editor: Data View & Variable View

The Data screen operates in two synchronized modes toggled at the top of the screen:

Mode A: Data View (Case Spreadsheet)

- Displays rows (cases/observations) and columns (variables).
- Edit a Cell: Tap any cell to open the edit dialog, enter a new numeric or string value, and tap Save. Changes immediately update the active dataset.
- Add / Delete Cases: Use '+ Case' in the toolbar to append a new row, or tap the trash icon next to a row number to remove that observation.
- Sync to Syntax: Tap 'Generate .sps' to convert the entire active dataset (dictionary + inline BEGIN DATA ... END DATA) into a reproducible PSPP syntax script.

Mode B: Variable View (Dictionary Metadata)

Tap any variable card in Variable View to configure its 8 standard SPSS/PSPP attributes:

- 1. Name: Variable identifier (e.g., baseline_bp, outcome_score).
- 2. Type: NUMERIC (F format) or STRING (A format).
- 3. Width & Decimals: Field width (e.g., 6) and decimal places (e.g., 1 for F6.1).
- 4. Label: Descriptive human-readable variable label printed in output tables.
- 5. Value Labels: Category mappings formatted as 1=Intensive, 2=Standard, 3=Placebo.
- 6. Missing Values: Discrete user-missing codes (e.g., -9, 999).
- 7. Measure Level: SCALE (continuous), ORDINAL (ordered), or NOMINAL (categorical).

5. Files, CSV Preview & .sav Binary Storage

- Open .sav / .zsav / .por / .csv: Tap 'Open Dataset' on the Files screen to pick a file via Android Storage Access Framework (SAF).
- CSV Import Preview: When opening a CSV/TSV file, PSPP Studio displays an interactive preview where you can switch delimiters (Comma, Semicolon, Tab, Pipe), toggle the header row, and inspect inferred column types (F vs A) before importing.
- Export Binary .sav: Tap 'Export .sav' to write a standards-compliant $FL2 binary system file (including long variable names and value labels) readable by desktop GNU PSPP and IBM SPSS.

6. Statistical Procedures Reference (Analyze Screen)

Select a procedure chip on the Analyze screen, pick your target variables, review or customize the generated PSPP syntax in the bottom preview box, and tap Run:

1. Descriptives (DESCRIPTIVES)

Computes N, Mean, Std. Deviation, Minimum, Maximum, Variance, Skewness, Kurtosis, and Std. Error of Mean for one or more continuous scale variables.

2. Frequencies (FREQUENCIES)

Produces frequency counts, percentages, valid/cumulative percentages, summary quantiles (Mean, Median, Mode, Std. Dev), and bar chart directives.

3. Crosstabs (CROSSTABS)

Builds two-way contingency tables (Row Variable BY Column Variable) with cell counts, row/column/total percentages, Pearson Chi-Square, Likelihood Ratio, and Phi.

4. Independent & Paired-Samples T-Test (T-TEST)

- Independent Mode: Compares means of test variables across two groups defined by a grouping variable (e.g., gender(1, 2) or group(1, 2)) with Levene's test and 95% CI.
- Paired Mode: Compares paired measurements (e.g., baseline_bp WITH followup_bp).

5. One-Way ANOVA (ONEWAY)

Tests differences in means across 2+ factor groups with between/within sums of squares, F-statistic, p-value, Levene homogeneity test, and Tukey/Bonferroni post-hoc.

6. Bivariate Correlation (CORRELATIONS)

Computes the Pearson r correlation matrix, two-tailed significance levels, and pairwise sample sizes across all selected numeric variables.

7. Linear Regression (REGRESSION)

Fits an Ordinary Least Squares (OLS) multiple linear regression model:

```
REGRESSION /VARIABLES=adherence_score baseline_bp age outcome_score
  /DEPENDENT=outcome_score /METHOD=ENTER /STATISTICS=COEFF R ANOVA CI.
```

8. Nonparametric Chi-Square (NPAR TESTS /CHISQUARE)

Evaluates goodness-of-fit against equal expected category proportions.

9. Scale Reliability (RELIABILITY /MODEL=ALPHA)

Computes Cronbach's Alpha internal consistency and item-total statistics.

7. Output Viewer, Exporting & Printing

After running syntax or an Analyze procedure, the Output screen displays the results:

- HTML Table View: Renders PSPP pivot tables with full light/dark styling and horizontal scrolling for wide multi-variable matrices.
- Plain Text & CSV Tabs: Switch view modes to inspect raw ASCII text tables or comma-separated values suitable for spreadsheet import.
- Export HTML / TXT / CSV: Saves the output file directly to device storage via SAF.
- Print / Save as PDF: Invokes the Android PrintManager adapter on the output WebView so you can save a paginated PDF report or send it to a wireless printer.
- Share Output: Opens the Android system Share Sheet to send results via email, messaging apps, or cloud drives.

8. Engine Mode Verification & Troubleshooting

- Verifying Native JNI Execution: Open the 'About' tab or check the top status badge. When libpspp_jni.so is loaded, the badge shows 'JNI Native Engine (libpspp_jni.so)' and no Approximation Mode warning banner is shown.
- Release Builds (-PallowFallback=false): Release APKs disable the Kotlin fallback evaluator entirely at build time, guaranteeing that 100% of statistical output is produced by the genuine C GNU PSPP 2.1.2 engine.
- Changing Language (EN / AR / FR): Tap the globe icon in the top app bar to switch between English, Arabic (with automatic Right-to-Left layout), and French.

License & Attribution

PSPP Studio embeds GNU PSPP 2.1.2 and is distributed under the GNU General Public License v3.0 or later (GPLv3+). Full license text and source patches are bundled inside the app under the About tab and in the repository's /patches directory.